

INDUSTRIAL PROGRAMMING LANGUAGE

MDA 3044

LAB 5

Name	KAUTHAR BINTI MOHD HISHAM
Matric No.	S76156
Program	BACHELOR OF SCIENCES (DATA ANALYTICS)
Group	K1

Question 3

```java
package lab5;

public class Q3 {
    public static void main(String[] args) {

        System.out.printf("%-12s%8s\n", "Miles", "Kilometers");

        for (int i = 1; i < 200; i++) {
            System.out.printf("%-12d%8.1f\n", i, i * 1.609);
        }
    }

}
```

```
Miles       Kilometers
1                1.6
2                3.2
3                4.8
4                6.4
5                8.0
6                9.7
7               11.3
8               12.9
9               14.5
10              16.1
11              17.7
12              19.3
13              20.9
14              22.5
15              24.1
185            297.7
186            299.3
187            300.9
188            302.5
189            304.1
190            305.7
191            307.3
192            308.9
193            310.5
194            312.1
195            313.8
196            315.4
197            317.0
198            318.6
199            320.2
```

Question 5

```java
package lab5;

public class Q5 {
    public static void main(String[] args) {
            int count = 0;

            for (int i = 100; i <= 1000; i++) {
                if (i % 5 == 0 && i % 6 == 0) {
                    System.out.printf(i + " ");
                    count++;
                    if (count % 10 == 0) {
                        System.out.println();
                    }
                }
            }
            if (count % 10 != 0) {
                System.out.println();
            }
    }
}
```

```
120 150 180 210 240 270 300 330 360 390
420 450 480 510 540 570 600 630 660 690
720 750 780 810 840 870 900 930 960 990
```

Question 6

```java
package lab5;

public class Q6 {
    public static void main(String[] args) {
            int count = 0;

            for (int i = 100; i <= 200; i++) {
                if ((i % 5 == 0 || i % 6 == 0) && !(i % 5 == 0 && i % 6 == 0)) {
                    System.out.print(i + " ");
                    count++;
                    if (count % 10 == 0) {
                        System.out.println();
                    }
                }
            }

            if (count % 10 != 0 && count > 0) {
                System.out.println();
            }
    }
}
```

```
100 102 105 108 110 114 115 125 126 130
132 135 138 140 144 145 155 156 160 162
165 168 170 174 175 185 186 190 192 195
198 200
```

Question 8

```java
package lab5;

public class Q8 {
    public static void main(String[] args) {

        int n = 0;

        while(Math.pow(n, 3) < 12000) {
            n++;
        }

        int N = n - 1;

        System.out.println(N + " is the largest number, such that " + N + "^3 is less than 12000");
        System.out.println("Proof: " + N + "^3 = " + Math.pow(N, 3));
        System.out.println("Proof: " + (N + 1) + "^3 = " + Math.pow(N + 1, 3));
    }
}
```

```
22 is the largest number, such that 22^3 is less than 12000
Proof: 22^3 = 10648.0
Proof: 23^3 = 12167.0
```

Question 10

```java
package lab5;

import java.util.Scanner;
public class Q10 {

    public static void main(String[] args) {
            Scanner input = new Scanner(System.in);

            System.out.print("Enter an integer: ");
            int number = input.nextInt();

            System.out.print("The smallest factors of " + number + " are: ");

            if (number <= 1) {
                System.out.println(number);
                input.close();
                return;
            }

            int divisor = 2;
            while (number > 1) {
                if (number % divisor == 0) {
                    System.out.print(divisor + (number == divisor ? "" : ", "));
                    number /= divisor;
                } else {
                    divisor++;
                }
            }

            System.out.println();
            input.close();

}
}
```

```
Enter an integer: 75
The smallest factors of 75 are: 3, 5, 5
```

Question 11

```java
package lab5;

public class Q11 {
 public static void main(String[] args ) {
  final int prime_numbers_per_line = 8;
  final int upperLimit = 1000;
  int count = 0;

  for (int number = 2; number <= upperLimit; number++) {
   if(isPrime(number)) {
    System.out.print(number + " ");
    count++;
    if (count % prime_numbers_per_line == 0 ) {
     System.out.println();
    }
   }
  }
 }

 public static boolean isPrime(int number) {
  for(int divisor = 2; divisor <= Math.sqrt(number); divisor++) {
   if (number % divisor == 0) {
    return false;
   }
  }
  return true;
 }

}
```

```
2 3 5 7 11 13 17 19
23 29 31 37 41 43 47 53
59 61 67 71 73 79 83 89
97 101 103 107 109 113 127 131
137 139 149 151 157 163 167 173
179 181 191 193 197 199 211 223
227 229 233 239 241 251 257 263
269 271 277 281 283 293 307 311
313 317 331 337 347 349 353 359
367 373 379 383 389 397 401 409
419 421 431 433 439 443 449 457
461 463 467 479 487 491 499 503
509 521 523 541 547 557 563 569
571 577 587 593 599 601 607 613
617 619 631 641 643 647 653 659
661 673 677 683 691 701 709 719
727 733 739 743 751 757 761 769
773 787 797 809 811 821 823 827
829 839 853 857 859 863 877 881
883 887 907 911 919 929 937 941
947 953 967 971 977 983 991 997
```

Question 12

```java
package lab5;

public class Q12 {
 public static void main(String[] args) {
  double sum = 0.0;

  for (int i = 1; i <= 20; i++) {
   sum += (double)i/(i + 2);
  }
  System.out.printf("The sum of the series is : %.4f\n", sum);
 }
}
```

```
The sum of the series is : 15.6184
```

Question 14

```java
package lab5;

public class Q14 {
 public static void main(String[] args) {
  double product = 1.0;

  for (int i = 1; i <= 625; i++) {
   product *= Math.sqrt((double)(i + 1) / (i + 2));
  }
  System.out.printf("The value of the product is : %.5f\n", product);
 }

}
```

```
The value of the product is : 0.05648
```

Question 15

```java
package lab5;

import java.util.Random;

public class Q15 {
 public static void main(String[] args) {
  int heads = 0;
  int tails = 0;
  int numOfFlips = 1000000;

  Random random = new Random();

  for (int i = 0; i < numOfFlips; i++) {
   if (random.nextBoolean()) {
    heads++;
   } else {
    tails++;
   }
  }

  System.out.println("Number of heads: " + heads);
  System.out.println("Number of tails: " + tails);
 }
}
```

```
Number of heads: 499253
Number of tails: 500747
```